PFP XI: Recreating Symbol Game Demonstrations with Source and Replay Custody

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Abstract

The Symbol Game demonstration corpus contains executable examples, retained model outputs, reference solutions, certificate checkers, informal mechanism sketches, and subsequently corrected findings. These objects support different claims. This paper reconstructs the corpus through a source-bound inventory and explicit replay procedures. Every result identifies the original source, the executed algorithm, its dependencies, and the obligation that its output checks. Historical reference self-scores and prompts containing answers remain visible as limitations. A copied receipt does not become a new experiment, and an unavailable operation does not become a verified computation. The present draft retains unresolved reconstruction obligations until the complete corpus and its positive examples have been independently replayed.

The reconstruction problem

A heterogeneous demonstration archive cannot be reproduced by rerunning one entry point. Some original modules refer to retired namespaces. Others call native certificate engines, include retained answers, or describe a mechanism without supplying executable cases. The findings also contain corrections that withdraw earlier conclusions. Reconstruction must retain the mathematical task and the historical record while making dependencies and interpretation explicit.

We use two source collections: the twenty findings notes and the complete tracked PFP demos subtree. The source census at the captured repository revision contains 8,037 files in the latter and twenty notes in the former, totaling 34,722,866 bytes. This census establishes the intended copy inventory. It does not establish that copy custody or execution has completed. The companion custody and execution records determine those outcomes separately.

Mechanism inventory

Mechanism	Input and output	Evidential scope
Findings inventory	Original source spans to named demonstration families	Coverage of the inspected notes
Raw source custody	Tracked source files to byte-preserved copies and SHA-256 bindings	File identity and inventory completeness
Archived algorithm replay	Original finite cases to original solver and checker outcomes	The pinned algorithm on those finite inputs
Native certificate replay	Exact rational inputs, executable digests, and candidate witnesses	The declared native checking obligations
Import adaptation	Retired import names to explicit copied or pinned modules	Namespace relocation only; mathematical bytes remain bound
Historical recount	Retained prompts, answers, labels, and corrections	Recount of retained observations rather than a new model trial

Mechanism	Input and output	Evidential scope
Package verification	Files, controls, manifests, and deterministic archive	Transport and replay custody rather than scientific truth

This paper assigns no authorization and changes no jurisdiction. It carries no operational authority or institutional standing.

Source identity and execution identity

The source snapshot records a Git revision, each tracked path, its Git blob identity, its raw file digest, and its destination digest. Raw digests are needed because parsed JSON and normalized line endings do not bind physical file bytes. The copied inventory must be a bijection with the declared tracked inventory. Additional, missing, redirected, or corrupted entries cause custody verification to fail. A retained historical file is inert evidence until a separately declared replay executes it.

An execution record additionally binds the algorithm and any adaptation, actual input and output bytes, process exit status, runtime configuration, and dependencies. Archived Python algorithms are described as source recreations. Native results identify the Rust executable and its independently checked certificate. Neither kind of record is presented as a new model-performance result.

Preserving the declared task

Changing an import path can restore executability without changing the intended computation. Such an adaptation is acceptable only when its permitted change is recorded and the original and adapted bytes remain available. Changes to carrier, case data, protected constraints, answer format, or checker semantics require separate disclosure. Success under an easier replacement does not reproduce the original demonstration.

For a rational graph example, the host minimizes a declared residual objective

$$\Phi_1(\omega) = \min_{u} \sum_{e=(a,b)} \left|\omega_e - (u_b - u_a)\right| .$$

Equality of primal and dual objectives is meaningful only after feasibility, divergence balance, and the dual box bounds have been checked. A scalar returned by a runner supplies no independent certificate. Higher-degree, gauge, stabilizer, and formal-mathematics demonstrations have their own carriers and obligations; the graph objective cannot silently substitute for those checks.

Historical findings and corrected interpretations

The preserved findings report reference scores and empirical observations together. Their correction notices impose material limits. Labels called `PASS_TO_MASTER` are reference self-scores. Several formal-mathematics panels used answer-bearing case identifiers. The reported paired-bout ten-of-ten count measured a field that was true in every retained record. Passing blind-cluster runs followed prompts containing the answer. The claimed twenty-eight-probe study has no retained study behind it. These statements must be analyzed as corrected historical records.

The stealth-model observations also lack a disclosed model identity, size, and training provenance. They fall outside the collection's declared open-weight challenge regime. Replaying a checker over a retained answer may reproduce a finite checking outcome, but it does not recover the missing live experiment or establish unaided model accuracy.

Genesis labels require particular care: the findings quarantine an always-passing artifact verifier and report timeout, failure, and absent-verifier cases among records previously labeled checked. A new finite example could demonstrate a mechanism, but it would be a new reconstruction with its own evidence, not a retroactive confirmation of those labels.

Replay protocol

1. Verify source custody and the pinned interpretation before execution.
2. Resolve explicit dependencies and validate executable digests.
3. Execute original finite cases with bounded resources and isolated outputs.
4. Preserve actual outputs and failures; run the corresponding checking obligation.
5. Rehash copied sources and controls after every mutable subprocess.
6. Emit a readable account and machine-readable records with the same identities.
7. Keep positive reconstruction gaps open until their computations are reproduced.

A refusal can correctly demonstrate a boundary when refusal is the declared example. It cannot satisfy a positive computation merely because a current API lacks that computation. Missing historical data and unavailable positive ports remain distinct reconstruction obligations.

Current evidence and remaining work

The source census is complete and the native host and certificate binaries have been rebuilt from the synced repository revision. Twenty focused synthetic custody controls exercise byte changes, missing and additional copies, redirected files, malformed manifests, and post-execution drift. These checks establish the behavior of custody machinery on their fixtures. They do not establish custody of the complete source snapshot or completion of every demonstration. Seventeen additional package-custody controls check source inclusion, traversal failure, path collisions, redirected entries, inventory drift, and archive-envelope drift. The draft PDF has also been rebuilt without clipping or missing-character warnings.

The completed findings inventory covers sixty challenge IDs, twenty-five Genesis IDs, and thirteen aggregate demonstration families. An independent replay checks the twenty note digests, 769 physical section and line bindings, and 343 referenced source files. The reproduction catalog additionally maps thirty-one dispatcher entries. These are coverage and source-identity results, not execution totals.

An initial C017 archived-algorithm replay executes sixty cases: fifty-seven are accepted by both retained checkers and three produce reference abstentions. Twelve declared mutations are rejected by both. The record remains a partial reconstruction with a reference residual; checker agreement alone does not prove algorithmic independence or recreate the unresolved positive obligations.

Eleven further finite fixtures exercise attention masking, activation patching, grammar-constrained emission, graph scratch operations, cache retention, residual steering, paired counting, deduplicated sampling, and packet/cemetery display. Their neural components use explicit CPU tensors rather than trained model runs. The native graph components retain exact requests, responses, and

executable digests. Twenty focused mechanism tests ran with explicit native capabilities. The planned empirical model studies remain unperformed.

One fixture exposes a concrete verification failure. A retained consistency checker accepts a schema-valid contradiction verdict for a graph whose native residual is zero. The schema therefore does not supply a mathematical obstruction witness. Another fixture emits the directed-cycle candidate `(1, 1, -2)` and checks its zero residual and the original cycle's exact optimal repair cost of three. Corrupted candidate and dual bytes fail native checking. These are finite component results, with no model-performance interpretation.

The MUB6–Regge prose example still lacks a declared optimization/geometry input and corresponding certifiers. Its unresolved record identifies those obligations; it is excluded from the eleven executed fixtures.

The separate retained MUB catalog now has a fresh native replay. Its seven exact triples and twenty-one transition matrices reproduce the original report. The Rust adaptation changes only the report's output directory; the mathematical owner and fixed catalog inputs remain unchanged. Two native controls establish the declared twenty sign patterns for each identity and distinguish a Fourier matrix's zero failures from the Tao matrix's twenty failures. Eleven custody controls check the output binding and reject corrupted or rehashed invalid reports. This is a finite catalog calculation, separately scoped from the symbolic affine ansatz and the unresolved geometry example.

Git history also supplies the twenty-five original Genesis specifications, harness, master report, and receipt. All twenty-eight recovered files are bound to their original Git blobs and raw digests. Replaying the original artifact verifier with empty, wrong-task, and unproved responses reproduces seventy-five erroneous `verified: true` outputs. Among eleven historical `CHECKED` labels, the report records six compilation timeouts, two failures, and three absent snippets; it records no successful compilation. Raw responses and Lean snippets were not persisted. The harness defect is reproduced, while the specifications' positive mathematical obligations are not established by that historical report. The recovered receipt also has a different identifier from the receipt cited in the findings note; their identity cannot be inferred from matching prose.

A fresh G07 computation now reproduces the complete original provider-free branch-fan census at its declared source pins. The two depth-fifteen trees have 963 and 1,847 nodes and reconcile the original 236 and 422 expanded tie counts. The source and native owners agree on 3,617 offered-challenge width cases; a separate native replay accounts for all thirty-five quiet histories, where width is undefined. All forty-four original held-outs are scored. The largest observed width is eleven. The proposed global bound remains an OPEN conjecture: the finite trees and original depth-sixty trajectories do not prove it. Six corrupted responses are rejected. This closes the mathematical baseline, while the five model arms and their comparative statistics remain unperformed.

The fresh G01 reconstruction proves the original reading-F targets F-1, F-5 and F-K under the unchanged sealed definitions. It checks eight explicitly reconstructed held-out statements, eleven target pins and ninety actual theorem and auxiliary axiom reports within the original whitelist. Seven original mutation controls are refused while a valid candidate remains accepted. The separately retained lexical supplement also rejects forbidden declarations in comments, closing a gap in the original guard without rewriting its receipts. The complete supplied two-item finite populations agree between the original Python and native owners on every candidate and class membership: F has ten responses in five classes; L has six in three; both arity-cost populations have two in one. Different refinement labels remain recorded and are checked against the shared full-permutation partition. The historical frozen judge packet was unavailable; the new held-out signatures are therefore

disclosed as reconstructions.

G02 now proves the original general cut-census formula $\mathrm{cutCensus}(n, a, b) = 2^{n-2}$ for distinct designated pairs within range, retaining its natural-subtraction guards. A clean Lean build checks thirteen original target and held-out pins; all fifty-two theorem and auxiliary reports obey the unchanged axiom whitelist. The unchanged Python sublist enumerator and a declared native bitmask checker agree on all 1,014 pair cases through carrier twelve. The original five mutation controls are refused and a known valid candidate passes. Source-byte screening detects forbidden comment and namespace tokens even when the kernel alone accepts them. An independent review's generated-input custody finding is closed by a separate mutation-only replay that hashes each compiler input before and after use. These formal results establish the definitions' mathematical properties without recreating the unperformed model-budget and external estate experiments.

G03 now executes every original chain-pair obligation for a newly declared, preregistered collapse-preserving adapter. Its explicit anchor structure selects no reading and remains separate from the withdrawn historical map. The original 455 pairs produce 1,820 distinct source calls; held-out chains add 1,444. The native checker evaluates all 9,792 base, renaming and reversal presentations. Independent replay validates 16,092 data/form bijections and every raw candidate and class-member correspondence. Both preregistered fresh-mark predicates are false at $(3, 1, 2)$ in all four reading/measure combinations. Existing unary forms allow cost-one repairs, so the two-item fresh mark is not minimal. This is a finite refutation for that adapter; the general G4 conjecture remains open. The original hostile controls are refused, including an adapter that preserves three collapsed pairs while breaking four chain-pair correspondences. Nineteen focused tests pass. Lossless evidence packaging and the original model and external estate experiments remain separate obligations.

The fresh G12 baseline compiles five original isomorphism constructions as faithful Prop wrappers and closed instances at the original concrete regime. Two actual environment censuses agree on twelve parked modules, 141 compiled constants and five constructors. Lean 4.33.0 reports only `propext` and `Quot.sound` for every wrapper and instance. The unchanged parked census and whole axiom-artifact gates pass; all forty-eight original parked reports remain unchanged. Five executed controls reject vacuous hypotheses, altered statements, unbound text censuses, misplaced wrappers and the original forbidden-token case. These source-bound checks establish the declared provider-free baseline. They do not recreate missing model responses or the broader external estate workflow.

An additional native audit checks all supplied rational boundary matrices in the eight original dossiers for demos 5, 6, 7 and 9. Every consecutive boundary product vanishes. Exact native ranks confirm the declared rational Betti numbers for both tetrahedral dossiers, both degree-three dossiers and the hollow five-simplex. They also expose concrete source errors. Demo 5's filled planar and Kagome complexes both have first Betti number zero; the stated values two and four count cycles in the underlying graphs. Their separately verified graph LP objectives remain valid edge-quotient calculations. Demo 9's claimed $S^2 \times S^2$ carrier instead yields Betti vector $(1, 0, 1, 2, 1)$ from its actual matrices, contradicting its stated second and third Betti numbers. Four native controls detect an invalid boundary product, refuse floating and ragged maps, and detect a false Betti claim. Integral torsion and manifold identity are separate obligations; rational ranks do not certify them.

A separate integer replay now checks the same eight complete carriers. It transposes the declared cochain maps into chain boundaries and retains twenty-two actual Smith certificates. Native replay checks each basis change and its unimodularity against the original boundary. The original torsion carrier has integer chain $H_2 = \mathbb{Z}/2$, while its mod-two dimensions are $(1, 0, 1, 1)$. Native integral and mod-two computations agree under the universal-coefficient dimension check in every degree

of all eight carriers. Four malformed-carrier and five damaged-certificate controls are refused. This closes the supplied carrier's torsion computation while leaving manifold identity and cup-product interpretations unestablished.

The full formal-lemma replay now executes all sixty-four retained cases and forty-eight mutations against the pinned Lean kernel. Thirty-six proposed proofs and twenty-two refutations are accepted. Four malformed statements receive warranted refusals; two reference proposals abstain after exhausting their decision-tactic heartbeat budget. All forty-eight mutations are rejected by both retained checkers. Those checkers share the same compiler wrapper, so agreement does not establish independent provenance. The execution tape retains the actual submission, judge, and generated module bytes as well as the kernel outputs. The original solver's two abstentions remain separate from any later proposals. Two supplemental candidates now close those particular targets: a refutation uses $a = b = 39$ for the false bounded-sum statement, and a proof derives $k \leq 49$ before bounding k^2 by $49^2 < 2600$. Both pass the unchanged kernel and checkers, and four altered-proof or wrong-kind controls are rejected. This two-target result remains separate from the original sixty-four-case solver run and its forty-eight mutations. The original transport retained decoded text observations; the supplemental transport additionally retains physical native output bytes before presenting the original text interface.

The original two Phi-two dossiers now also replay through a scoped native composition: their objectives are one and two with zero gap, and all eight retained near-miss certificates are refused. This composition does not repair the production adapter's separate missing-port diagnostic. The four original degree-three and degree-four witnesses also pass the existing native verifiers: their objectives are respectively one, zero, one and two, each with zero gap. All sixteen original near-misses are refused. The exact source Bianchi and declared cochain-pairing computations remain separate from these rational certificates and the missing manifold/cup-product obligations. Three further original topology repairs preserve their full declared carriers and weights: asymmetric K4 has native defect twenty-one, the four-spoke wheel sixteen and K5 twenty-seven; each supplied repaired state has defect zero. Exact potential, idempotence and damaged-candidate controls are retained. These computations check supplied repairs; scripted proposer loops do not measure model discovery.

The original four-stage AST source progression now also has a closed native replay. Its defect pairs are $(7,7), (0,0), (0,0), (0,0)$, while active defect counts are two, two, one and zero. Only the fourth stage is exact. All four certificates replay; damaged metrics, changed states and a broken chain-law carrier are refused. The stages have different carriers, so these values do not establish descent on one fixed carrier. This original executable fixture remains separate from the three historical manually scored sessions whose exact states are unavailable.

Bounded source recovery also reproduces the original Demo 2 restricted TFI gate. It checks all sixteen entries of the declared two-site matrix and all twenty-three original controls. Its result is `RestrictedSignGaugeRepair`, with no native Phi certificate. The recovered R2 carrier is the open four-site chain; a previous ring fixture cannot reproduce it. Its retained seven-move response remains `WITNESS_UNCHECKED`. Exact recovered source assignments now reproduce all 120 role texts across fifteen historical rounds. They expose synthetic scoring and the original false AST terminal: a two-step run announces exact repair at defect pair $(4,1)$ with four active defects. Source/template reproduction establishes that failure mechanism while the original physical representations, native carriers and live-call authenticity remain unresolved.

Other closed replays give concrete witnesses. A four-vertex graph with edges `01,02,13,23` reaches

the target `03,12,23` through the local-complementation sequence `[0,2,3]`. Both retained simulations check that sequence. Of forty cases, thirty-eight yield accepted non-refusal outputs and two yield accepted scope refusals; all thirty-nine retained mutations are rejected by both checkers. A triangle is three-colorable with witness `[0,1,2]`; its two-color encoding instead has a checked finite DRAT refutation. The SAT and DRAT branches belong to one evaluator. Three malformed-input boundary disagreements remain visible in its sixty-four-case run, despite all forty-eight mutations being rejected. For the finite poset category `0`→`1`→`2`, the product of objects `1` and `2` has apex `1` with legs `id1` and `f_1_2`. Both retained checkers accept that witness, but disagree on thirty-one of forty-eight hostile mutations across the full category challenge. A passing example therefore does not certify the checker family's complete soundness.

The complete copy snapshot and the full fresh recreation results remain in progress. The companion requirements document defines the completion audit. The current version is an author-review draft and makes no complete-corpus recreation claim until that audit has actual supporting evidence.

Reproducibility and limits

The companion package retains source and adapted runtime identities separately. Its code supplies reproduction machinery; dated output receipts supply execution evidence. Package tests run from the extracted paper directory. Native operations require explicitly supplied, digest-checked executables and a pinned engine capability where declared. Missing configuration yields an explicit residual.

No universal natural-language formulation guarantee, unrestricted solver, general model-accuracy result, solution of the dimension-six mutually unbiased bases problem, or proof of every historical narrative follows from these finite checks. The goal is a faithful and inspectable reconstruction of the declared corpus.

References

The companion source inventory and references identify the twenty original findings notes, the complete PFP source tree, corrected Papers III, VI, VIII, IX, and X, and each executed engine or retained algorithm. These local artifacts are primary evidence for the corpus reconstruction; no performance guarantee is imported from unrelated literature.

Appendix: Tasks retained by the reconstruction

The following inventory keeps every named challenge and aggregate family in the reconstruction. Each row names an original task and its checking obligation. It is a scope register, not a table of completed computations. The companion source-bound inventory supplies the original note spans and source digests; closed replay records supply the separate execution outcomes.

Mechanism: C001 to C010

Task	Original demonstration	Required computation or check
C001	Distractor-Key Attention Masking at Task Level	Replay the attention-hook fixture, empty-mask token identity and task checker; distinguish actual neural inference from fixtures.

Task	Original demonstration	Required computation or check
C002	Carrier Attention Steering Chosen on Public, Frozen for Hidden	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.
C003	Head Attribution by Activation Patching, Then Targeted Boost	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.
C004	Carrier Mask vs Grammar Mask vs Free Decoding Across a Model Ladder	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.
C005	Hosted Decoding Controls Bake-off	Replay all retained four-arm responses with exact potential propagation and recount paired denominators.
C006	Host-Maintained Scratch Space vs Chain of Thought	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.
C007	Context Compaction and KV-Prefix Surgery over Long Horizons	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.
C008	Activation Steering for Sign Discipline	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.
C009	Model-Size Ladder: Where Does the Game Help Most	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.
C010	Repeated-Sampling Frontier and Cost per Verified Success	Record declared actual model, intervention, request and output bytes separately from independent graph correctness.

Cohomology: C011 to C020

Task	Original demonstration	Required computation or check
C011	Cycle-Rank Ladder: Minimum-L1 Coboundary Repair	Verify exact potential/residual and feasible dual circulation for minimum weighted L1 repair.
C012	Cellular Sheaf Cohomology H^0 and H^1	Verify cellular-sheaf restriction maps, H0/H1 bases and ranks.
C013	Integral Homology with Torsion and Smith Normal Form Certificates	Verify Smith-normal-form transforms and integral homology/torsion certificates.
C014	Exact Discrete Hodge Decomposition	Verify exact/coexact/harmonic decomposition with declared inner product.
C015	Persistent Homology Barcodes with Reduction Certificates	Verify F2 filtration-preserving reduction and persistence certificates.
C016	Cohomology Rings: Telling Spaces Apart with Cup Products	Verify cocycle, quotient and cup-product relations.
C017	Nonabelian Lattice Holonomy and Flatness	Replay every S3/Q8 plaquette, pure-gauge transformation, torus loop and retained model claim.
C018	Frustration Index of Signed Graphs	Replay each declared exact algebraic certificate and all carrier, restriction-map and closure obligations.
C019	Degree-2 Obstructions at Scale	Replay declared DQ maps, exact ranks and degree-2 extension/obstruction witness.
C020	Spectral Sequence Pages of a Double Complex	Replay each declared exact algebraic certificate and all carrier, restriction-map and closure obligations.

Quantum: C021 to C030

Task	Original demonstration	Required computation or check
C021	Q40-R: The ChatGPT Stabilizer Challenge Run as Specified	Verify commutation, Pauli phases, exact energy and degeneracy.
C022	Stabilizer Code Distance with Lower-Bound Certificates	Verify actual logical witness and distance lower-bound certificate.
C023	Clifford Circuit Synthesis from Tableaux	Replay each declared finite-field/cyclotomic certificate, phases and structural constraints.
C024	Local-Clifford Equivalence of Graph States	Replay each declared finite-field/cyclotomic certificate, phases and structural constraints.
C025	Exact Mutually Unbiased Bases in Prime-Power Dimensions	Replay each declared finite-field/cyclotomic certificate, phases and structural constraints.
C026	Frustration-Free Projector Hamiltonians	Replay each declared finite-field/cyclotomic certificate, phases and structural constraints.
C027	Kochen-Specker Sets and Contextuality Certificates	Check ray coloring or actual uncolorability refutation.
C028	Fermion-to-Qubit Mappings with Exact CAR Checks	Replay each declared finite-field/cyclotomic certificate, phases and structural constraints.
C029	Minimum-Weight Surface-Code Decoding	Replay each declared finite-field/cyclotomic certificate, phases and structural constraints.
C030	Tensor-Network Contraction Order Under a Cost Bound	Replay each declared finite-field/cyclotomic certificate, phases and structural constraints.

Physics: C031 to C040

Task	Original demonstration	Required computation or check
C031	Anomaly-Free Chiral U(1) Charge Assignments	Verify linear/cubic charge sums, chirality/primitiveness and bounded completion/count obligations.
C032	Six-Dimensional Supergravity Anomaly Cancellation	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C033	Weight Multiplicities of Simple Lie Algebras	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C034	Calabi-Yau Hodge Numbers from Reflexive Polytopes	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C035	Mathieu Moonshine from the K3 Elliptic Genus	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C036	Genus-Zero Invariants of the Quintic via Mirror Symmetry	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C037	Fusion Rules from the Verlinde Formula	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C038	Combinatorial Curvature and Discrete Gauss-Bonnet	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C039	Lattice Gauge Theory: Gauss Law and Topological Degeneracy	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.
C040	M-Theory on Tori: Cremmer-Julia Scalar Counting	Replay the concrete finite algebraic counts or transport identities against unchanged declared inputs.

Formal mathematics: C041 to C050

Task	Original demonstration	Required computation or check
C041	Lean 4 Proofs of Hidden Finite Lemmas	Compile actual proof bytes using pinned Lean and inspect axioms separately from UNKNOWN status.
C042	SAT Encodings with DRAT and LRAT Certificates	Replay CNF assignment or actual DRAT/LRAT proof.
C043	Ramsey and van der Waerden Lower-Bound Constructions	Check all fixed-color constraints and monochromatic obstructions or actual UNSAT proof.
C044	Finite Group Identification with Isomorphism Witnesses	Check group axioms, explicit bijective homomorphism or invariant separation.
C045	Knot Polynomials from Planar-Diagram Codes	Recompute PD validity and normalized exact knot invariants independently of refusal.
C046	Knuth-Bendix Completion with Joinability Certificates	Check termination order, original axiom coverage and all critical-pair joinability.
C047	Recurrence and Generating-Function Discovery	Verify recurrence against prefix/held-out terms and nonzero absence minor.
C048	Ternary Quadratic Forms: Solution or Local Obstruction	Verify nonzero rational point or actual local Hilbert obstruction.
C049	Universal Constructions in Finite Categories	Verify universal factorization, natural unit/counit and triangle identities.
C050	Equational Theorem Proving with Checkable Rewrite Proofs	Replay every positioned substitution step or all finite countermodel axioms.

Program synthesis: C051 to C060

Task	Original demonstration	Required computation or check
C051	Exact LP Solver with Dual and Farkas Certificates	Check LP primal/dual/objective, Farkas feasibility or unbounded ray.
C052	Max-Flow with Min-Cut Certificates	Check per-arc capacities, conservation, cut partition and exact flow-cut equality.
C053	Bareiss Fraction-Free Matrix Inversion	Verify determinant/adjugate/inverse identities or nonzero kernel vector.
C054	Exact Linear Systems with Inconsistency Witnesses	Check Ax=b, independent null basis and rank minors or yTA=0 and yTb!=0.
C055	Assignment Problems with Dual Potentials	Verify allowed permutation, objective, dual slackness or actual Hall violation.
C056	DFA Minimization and Equivalence with Distinguishing Strings	Verify complete DFA transitions, access/separating words and bijection or counterword.
C057	Sorting Network Synthesis	Check prefix, forbidden comparators and all binary input vectors.
C058	Straight-Line Superoptimization	Replay SSA, width and length and distinguish exhaustive-small from sampled-wide equivalence.
C059	SQL Synthesis under Three-Valued-Logic Fences	Replay SQL and independent relational bags including every protected-column preflight.
C060	Checker Synthesis Scored by Mutation Kill Rate	Execute the actual proposed checker program against valid/invalid certificates and hostile mutations.

Aggregate demonstrations

These families collect examples that the sixty challenge IDs do not cover. Historical traces and current computations retain separate denominators.

Demonstration family	Required recreation
Finite meron-cluster oracle and answer-leakage replay	Recompute 256 breakup graphs, 1296 configurations, 192 negative configurations, 199 zero-charge and 57 cancelling subgraphs, and Z=912; replay every retained five-arm observation and prompt-leak control.
Displayed two-phase affine ansatz cancellation	Replay gamma1 and gamma2 for every three-positive/three-negative sign pattern and the actual native fixture. Separate symbolic pattern count from historical sampled phase-point wording.
Three retained AST-refactoring sessions	Replay each three-session packet chain, actual descent obligations and final nonzero residual.
Three retained specimen-reduction sessions	Replay complete chains against original carrier; separate zero metrics, unresolved defect count and final exact-repair obligation.
Unified packet/trace census	Recount every available stream and transition; distinguish present files from the former 29-stream/107-transition census.
Eight-beat cognitive-auditing narratives	Recreate the declared per-round narrative/packet synchronization and audit all eight role observations without treating persona agreement as certificate.
Four-arm conditioning and retained K4 traces	Replay every retained baseline/hinted/partially-masked/scratchpad trace and current bound control/oracle with actual prompt, reply and certificate bytes.
Failure-detector/farm observations	Replay every retained typed failure and positive control; reproduce notice-driven rejection of the alleged 28-probe study.
Five named Hamiltonian systems and demo2 bouts	Recreate all five named finite systems and every available retained bout; separate exact cancellation, licensed conditional mechanism and open sign obstruction.
Ten-trial proposer/adversary recount	Recount actual target/challenge contingency table and separately replay deposited proposals through bounded certificate gate.
Diamond, holonomy3c, K4 rank-three and meron carriers	Recreate exact three-cycle, shared-edge diamond, K4 rank-three and two-state meron carriers, gauge invariance and global cycle checks; recount every retained probe.
61 retained detector candidates	Recount total/status/fixture custody against unchanged historical catalog; exercise valid/invalid candidates independently.
Cross-domain hostile-certificate matrix	Replay every actual mutation for the four cited families and current challenges; derive denominator from bound files and count checker crashes separately.

Genesis: G01 to G25

The recovered specifications preserve the tasks below. Their original harness falsifiers have been executed. Fresh G01, G02, G07 and G12 mathematical baselines and G03's newly declared adapter now have separate checked computations; their empirical arms remain unperformed. Each positive

reconstruction requires its declared definitions, sealed source pins, actual program or proof, and the original checking obligations. The saved historical response summaries do not contain the absent proof bytes.

Task	Original demonstration
G01	Reading F Non-Uniqueness in Lean
G02	The Cut Census Formula 2^(n-2)
G03	G4 as a Replayable Finite Check
G04	X-3: Is the First-Demand Index Isomorphism-Invariant
G05	X-1: A Decision Procedure for Demand
G06	NS-10: State-Change Counts on Commuting Squares
G07	Branch Fan Widths under mu2
G08	Alpha Engine: Live Injections for G10, G13 and G14
G09	Alpha Engine: Python Driver Refinement for One Fold
G10	The Symbol Game as an Alpha Engine Proposal Source
G11	Wiring the m114 Solver Adapter into the Search Policies
G12	Prop Wrappers for the Parked Isomorphism Modules
G13	Genesis Crosswalk: Binding the Three Native Ports
G14	OB-1: Searching for an Endogenous Demand Generator
G15	Unboundedness of Engine Driver Paths
G16	Tied-Minima Census Beyond Two Items and Arity Two
G17	Second Derivation: Challenges 3 to 5
G18	X-6: The Remaining Definability Census Rows
G19	NS-5: A Proof-Relevant Cell Calculus
G20	H3V7N5QF: Closed Forms for Shortest-Sequence Counts
G21	H3V7N5QF: A Reachability Criterion for C-R2
G22	H3V7N5QF: Lower Bounds for A-Perform Changes
G23	H3V7N5QF: Choice-Free Reproofs
G24	Seed-Edit Ablation Bench for the Four Holes
G25	Genesis as Omega Engine Foundation: A Checked Crosswalk

Executable dispatcher surface

The preserved runner also exposes the following thirty-one dispatch entries. They overlap the families above and therefore cannot be added to the challenge counts as independent experiments. Each entry has its own closed process record or an explicitly retained execution gap in the companion coverage records.

Dispatcher entry
`antimorphism_farm`
`demo_1`
`demo_10`
`demo_11`
`demo_12`
`demo_2`
`demo_3`
`demo_3_mub6`
`demo_4`
`demo_4e_cogsci_autopoiesis`
`demo_5`
`demo_6`

Dispatcher entry
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Higher-dimensional meron cluster demonstrations and empirical receipts

The preserved runner has been extended with five higher-dimensional topological meron demonstrations connected directly to the core Symbol Game execution interface (`symbol-game demo run`):

1. **2D Staggered Gross-Neveu Meron Cluster Solver (`demo_meron_2d_gross_neveu`)**: Operates on a discrete 2D torus T^2 (4×4 sites). Decomposes 4-site plaquettes into uncoupled clusters and performs orientation-reversing involution τ on free meron clusters ($N_M = 4$). Verifies exact partition sum $Z = 150$, exact pairwise cancellation of odd-winding merons (2 cancelled pairs), zero duality gap ($\Delta = 0$), and 4 wire dialogue turns adhering strictly to the 38-glyph alphabet Γ_{38}. Sealed receipt: `docs/live_openrouter_2026_10_03_meron/demo_meron_2d_gross_neveu/receipt.json` (`9636840c48f1b80ac61219ee94f228aa58ff972231b3e93c2b53f18989e288e4`).

2. **3D Meron String Linking and Winding Cancellation (`demo_meron_3d_string_linking`)**: Operates on a discrete 3D spacetime torus T^3 ($6 \times 6 \times 6$ sites). Evaluates 1D closed string loops with exact winding vectors $W(c) \in \mathbb{Z}^3$, topological winding parity $I(c) = (W_x + W_y + W_z) \bmod 2 \in \{0, 1\}$, and discrete Gauss linking numbers $Lk(c_1, c_2) \in \mathbb{Z}$ (distinguishing trivial unlinked pairs from $Lk = 1$ Hopf links). Verifies exact partition sum $Z = 310$, exact pairwise cancellation of 3 odd-winding string pairs under orientation flip τ, zero duality gap ($\Delta = 0$), and 4 wire dialogue turns in pure Γ_{38}. Sealed receipt: `docs/live_openrouter_2026_10_03_meron/demo_meron_3d_string_linking/receipt.json` (`de341fbc181fb0f9f4427b1012389c98d48c43071ff8ea84fecc17e30f05052e`).

3. **Offline mock of the 2D meron cluster model ladder (`demo_meron_2d_benchmark`)**: The sealed run sets live to false. Each turn is filled by `generate_mock_model_response`, which returns the expected wire. The Llama 3.1 zero-shot arm wraps that wire in a sentence, and the extractor still accepts it. The recorded 24/24 wire, opcode, and exact-match rates are that mock scorer. Its latencies are the constants 850 ms, 350 ms, and 120 ms. These figures are

not an OpenRouter measurement. A live invocation of this scenario is refused. Sealed receipt: `docs/live_openrouter_2026_10_03_meron/demo_meron_2d_benchmark/receipt.json` (669d755d782d9e9cfc48205441b1f787ff434df3ac24c0e69ae1227874419ea6).

4. **4D Half-Instanton Second Chern Sign Cancellation (`demo_meron_4d_half_instanton`)**: Operates on a discrete 4D spacetime torus T^4 ($4 \times 4 \times 4 \times 4$ sites, $V = 256$). Formulates localized half-instanton cluster centers $c \in T^4$ with topological chirality $q_c \in \{\pm 1/2\}$ and discrete topological charge density $q(x) = \sum_c q_c \delta(x, x_c)$, evaluating discrete second Chern numbers $c_2 = \sum_x q(x) = \frac{1}{2}(N_+ - N_-) \in \frac{1}{2}\mathbb{Z}$. The fermion determinant sign $\sigma(C) = (-1)^{2Q(C)} = (-1)^{N_+ - N_-}$ is negative for odd half-instanton sectors and cancels with the orientation-flipped partner $\sigma(\tau(C)) = +1$ under $\tau : (N_+, N_-) \mapsto (N_-, N_+)$, while zero-charge sectors ($N_+ = N_-$) evaluate to $\sigma(C) = +1$. The sealed run records partition sum $Z = 450$, cancellation of 3 odd half-instanton pairs, zero duality gap ($\Delta = 0$), and 4 wire dialogue turns in pure Γ_{38}. Sealed receipt: `docs/live_openrouter_2026_10_03_meron/demo_meron_4d_half_instanton/receipt.json` (7c20f3dcbb1995020718d250f90029ca97073fe8cc48546f90859392f416643a).

5. **4D SU(2) Gauge-Coupled Topological Cluster Solver (`demo_meron_4d_su2_gauge`)**: Couples the 4D topological cluster solver to non-abelian $SU(2)$ link holonomy variables represented via exact rational unit quaternions (`RationalSU2`) on larger lattices ($8 \times 8 \times 4 \times 4$ sites, $V = 1024$, 4096 links, 6144 plaquettes). Evaluates plaquette holonomies $U_{\mu\nu}(x)$, exact rational Wilson gauge action changes $\Delta S_W \in \mathbb{Q}$, and exact gauge invariance under local transformations $\Omega(x) \in SU(2)$. Evaluates discrete second Chern topological charge density $q(x) = \frac{1}{2}(P_{01} \cdot P_{23} - P_{02} \cdot P_{13} + P_{03} \cdot P_{12}) \in \frac{1}{2}\mathbb{Z}$ and total second Chern number $c_2 = \sum_x q(x) \in \frac{1}{2}\mathbb{Z}$. Demonstrates that orientation-reversing involution on the gauge field ($\tau : U_0 \mapsto U_0^\dagger$) inverts $c_2 \mapsto -c_2$ and induces exact pairwise cancellation of odd half-instanton sectors in the fermion determinant sign. Confirms exact partition sum $Z = 450$, exact cancellation of 3 odd half-instanton pairs, zero duality gap ($\Delta = 0$), and 4 wire dialogue turns adhering strictly to Γ_{38}. Sealed receipt: `docs/live_openrouter_2026_10_03_meron/demo_meron_4d_su2_gauge/receipt.json` (972e01277b53cfb0cbb466bf30ca4bedf66bb851a9d76b5e86976837447471ba).

All five receipts pass independent receipt verification (`verify()`), with status `VERIFIED_FIXTURE` and `reproduced: True`, and with zero float conversions. The benchmark rates above are the offline mock scorer.